
Golden-Section Search

Derivative-free optimization by interval reduction

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Four questions

- 1 Why does the constant $0.618\dots$ appear?
- 2 How can one function value be reused at every step?
- 3 What accuracy is guaranteed after a fixed budget?
- 4 When should we *not* trust the answer?

The working viewpoint

The method does not chase a derivative. It shrinks a bracket that is known to contain one optimum.

SECTION 1

Problem Setting

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SECTION 2

The Search Mechanism

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SECTION 3

Convergence and Cost

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Reliable Computation

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SECTION 5

Mathematical Problems

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Application Problems

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SECTION 7

Summary & References

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